Experiment 2:

Program Evaluation and Review Technique (PERT)

Aim

To develop a project schedule using the Program Evaluation and Review Technique (PERT) in ProjectLibre by estimating task durations, defining task dependencies, and analyzing the project completion time.

Algorithm

1. Start ProjectLibre.
2. Create a new project by entering the project name and start date.
3. Identify all project activities.
4. Estimate the optimistic (O), most likely (M), and pessimistic (P) durations for each activity.
5. Calculate the expected time for each activity using the PERT formula:
   Expected Time (TE) = (O + 4M + P) / 6
6. Enter the expected durations into ProjectLibre.
7. Define the dependencies between the activities.
8. Generate the project schedule and identify the critical path.
9. Save the project and review the Gantt Chart and Network Diagram.

Procedure

1. Open ProjectLibre.
2. Create a new project by selecting File → New.
3. Enter the project details such as project name and start date.
4. List all project activities.
5. Estimate the Optimistic, Most Likely, and Pessimistic times for each activity.
6. Calculate the expected duration (TE) for every activity.
7. Enter the calculated durations into the Gantt Chart.
8. Specify the predecessor relationships between tasks.
9. View the generated Gantt Chart and Network Diagram.

OUTPUT:



